SESSION 5 - REGULAR EXPRESSIONS

After this session, students are able to apply regular expressions for data validations with the assistances of attributes and reflections.

I. ATTRIBUTES

Attributes can be considered as techniques that allow changing behaviors of methods at runtime, performing compile time operations or handling unmanaged codes.

There are 2 types of attributes: built-in and custom.

In this session, attributes are considered as means containing regular expressions for data validations.

Create a custom attribute as followed:

```
[AttributeUsage(AttributeTargets.Field | AttributeTargets.Property,
AllowMultiple = true)]
public class EmailCheckAttribute: Attribute{
    public string RE{get; set;}
    public EmailCheckAttribute(string re){
        this.RE = re;
    }
}
```

Assume that you are required to create a model for staff information. Each staff is described as a staff id, staff name, email, phone number, photo, starting date.

Your staff model can be as:

```
public class Staff{
    public int StaffID{get; set;}
    public string StaffName{get; set;}
    [EmailCheck("^\\S+@\\S+\\.\\S+$")]
    public string Email{get; set;}
    public string PhoneNumber{get; set;}
    public DateTime StartingDate{get; set;}
    public string Photo { get; set; }
}
```

In ***Staff.cs***, the property **Email** is annotated as a custom attribute that has a regular expression to describe an email address should be.

II. REFLECTION

Reflection is a mechanism of discorvering class information solely at runtime. The **System.Reflection** is the root namespace that contains classes to implement the reflection.

The following code uses reflection to validate the email address of a user:

```
public bool CheckEmail(Staff staff){
        Type StaffType = staff.GetType();
        PropertyInfo propertyInfo = StaffType.GetProperty("Email");
        if (propertyInfo != null){
            EmailCheckAttribute customAttribute =
              propertyInfo.GetCustomAttribute<EmailCheckAttribute>();
            if (customAttribute != null){
                object value =
                            propertyInfo.GetValue(staff);
                Regex validateEmailRegex =
                            new Regex(customAttribute.RE);
                return validateEmailRegex.IsMatch(value.ToString());
            }
        }
        return true;
}
```

III. EXERCISES

1. Define an application that uses attributes, reflections in coordination with regular expressions for data validations.

   Hint

   - Use a readable project structure.

2. Give a regular expression to describe phone numbers in all the various forms (international codes, local numbers).

   Apply the regular expression to check a valid phone number of a staff.

3. Write a report about those.